

Run 05 | Business exit

Global (including India)

IV + controls + NES

Target Entrepreneurial/business exit in the past 12 months; binary 1 vs. 0
Sample 1,331,428 respondents **Coverage** 89 countries; 2015–2022
Positive cases 61,654 (4.63%) **Case time** 57m 52s
Train / val. / test 905,374 / 159,770 / 266,284 (68% / 12% / 20%, stratified)

SCT predictors (12): Self-efficacy, Opportunity perception, Fear of failure, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.

Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.

NES controls (12): Financing, Government support, Taxes and bureaucracy, Government programs, School entrepreneurship education, Higher/vocational entrepreneurship education, R&D transfer, Commercial/legal/professional services, Market changes, Market entry, Physical infrastructure, Cultural and social norms. Total predictors: 47.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.8000). Highest F1: Random Forest (0.2611). For XGBoost, adding NES changed ROC-AUC by +0.0002 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.8000	0.2066	89.73	19.23	38.05	0.2555	19s
Random Forest	0.7957	0.2198	92.18	23.20	29.85	0.2611	4m 30s
CatBoost	0.7936	0.2009	90.39	19.72	34.97	0.2522	2m 33s
H2O GBM	0.7931	0.1966	90.41	19.59	34.48	0.2499	1m 38s
Lasso	0.7724	0.1488	89.17	16.27	32.32	0.2165	13m 04s
Elastic Net	0.7724	0.1488	89.17	16.28	32.31	0.2165	11m 19s
Logistic Regression	0.7722	0.1482	89.17	16.30	32.37	0.2168	2m 16s
Ridge	0.7716	0.1474	88.41	15.88	34.97	0.2184	15m 48s
Naive Bayes	0.7006	0.0963	83.44	10.73	35.19	0.1645	4m 40s
Decision Tree	0.5000	0.0463	4.63	4.63	100.00	0.0885	34s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 6

Rank	Predictor	Raw importance	Share%
1	Age	2,799.28	11.42
2	Household size	1,790.98	7.31
3	Entrepreneurial intention	1,396.12	5.70
4	Occupational status	1,374.10	5.61
5	Business angel	1,084.30	4.42
6	Education attainment	1,065.48	4.35

Share is the percentage of total forest impurity importance across all 88 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	361.45	1.47
2	Prosperity index score	285.74	1.17

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Leading NES predictors within the same forest

Rank	NES predictor	Raw importance	Share%
1	Physical infrastructure	208.73	0.85
2	School entrepreneurship education	206.55	0.84
3	Cultural and social norms	202.93	0.83

NES shares use total forest importance.

Source: Completed Objective 5 NES extension batch, 29 September 2026; 01_obj5_global_iv_cv_nes_EXIT_ENT_harmonized. Scores and importance are taken directly from saved CSV outputs.